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By Amit Katiyar (MCA-JNU)



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TARGET- NIMCET / CUET.PG By: Amit Katiyar (MCA-JNU)

HEIGHT & DISTANCE

NIMCET PYQ



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1. A person stands at a point A due South of a tower and observes that its elevation is 60° . He then walks westwards towards B, where the elevation is 45° . At a point C on AB produced, he finds it to be 30° . then, $\frac{AB}{BC}$ is equal to
NIMCET 2010
(a) $\frac{1}{2}$ (b) 1 (c) 2 (d) $\frac{5}{2}$
2. If two towers of height h_1 and h_2 subtend angles 60° and 30° respectively at the mid-point of the line joining their feet, then $h_1 : h_2$ is
NIMCET 2012
(a) 1 : 2 (b) 1 : 3 (c) 2 : 1 (d) 3 : 1
3. A man observes the angle of elevation of the top of a mountain to be 30° . He walks 1000 ft nearer and finds the angle of elevation to be 45° . what is the distance of the first point of observation from the foot of the mountain?
NIMCET 2013
(a) $500\sqrt{3}(\sqrt{3} + 1)$ ft (b) $500(\sqrt{3} + 1)$ ft
(c) $500(\sqrt{3} - 1)$ ft (d) $\sqrt{3}(\sqrt{3} - 1)$ ft
4. From three collinear points A, B and C on a level ground which can be on the same side of a tower, the angles of elevation of the top of the tower are 30° , 45° and 60° , respectively. If $BC = 60$ m, then AB is
NIMCET 2014
(a) $15\sqrt{3}$ m (b) $30\sqrt{3}$ m
(c) $45\sqrt{3}$ m (d) $60\sqrt{3}$ m
5. Two towers face each other separated by a distance of 25 m. As seen from the top of the first tower, the angle of depression of the second tower's base is 60° and that of the top is 30° . The height (in meters) of the second tower is
NIMCET 2015
(a) $\frac{50}{\sqrt{3}}$ (b) $\frac{25}{\sqrt{3}}$ (c) 50 (d) $25\sqrt{3}$
6. In a right-angled triangle, the hypotenuse is four time the perpendicular drawn to it from the opposite vertex. The value of one of the acute angle is
NIMCET 2015
(a) 45° (b) 30° (c) 15° (d) None of these
7. A harbour lies in a direction 60° South of West from a fort and at a distance 30 km from it, a ship sets out from the harbour at noon and sails due East at 10 km an hour. The time at which the ship will be 70 km from the fort is
NIMCET 2015
(a) 7 pm (b) 8 pm (c) 5 pm (d) 10 pm

8. A circus artist is climbing a 20 m long rope, which is tightly stretched and tied from the top of a vertical pole to the ground. Find the height of the pole, if the angle made by the rope with the ground level is 30°
NIMCET 2016
(a) 10 m (b) 30 m
(c) 20 m (d) 40 m
9. Angle of elevation of the top tower from 3 points (collinear) A, B and C on a road leading to the foot of the tower are 30° , 45° and 60° , respectively. The ratio of AB and BC is
NIMCET 2020
(a) $\sqrt{3} : 1$ (b) $\sqrt{3} : 2$
(c) 1 : 2 (d) $2 : \sqrt{3}$
10. Angle of elevation of the top tower from 3 (collinear) A, B and C on a road leading to the foot of the tower are 30° , 45° and 60° , respectively. The ratio of AB and BC is
NIMCET 2022
(a) $\sqrt{3} : 1$ (b) $\sqrt{3} : 2$
(c) 1 : 2 (d) $2 : \sqrt{3}$
11. If the angle of elevation of the top of a hill from each of the vertices A, B and C a horizontal is α , then the height of the hill is
NIMCET 2022
(a) $\frac{1}{2} b \tan \alpha \sec \beta$ (b) $\frac{1}{2} b \tan \alpha \csc A$
(c) $\frac{1}{2} c \tan \alpha \sin C$ (d) $\frac{1}{2} \alpha \tan \alpha \csc A$
12. A line segment AB of length 10 meters is passing through the foot of the perpendicular of a pillar, which is standing at right angle to the ground. Top of the pillar subtends angles $\tan^{-1} 3$ and $\tan^{-1} 2$ at A and B respectively. Which of the following choice represents the height of the pillar?
NIMCET 2023
(a) 10 meters (b) 8 meters
(c) 12 meters (d) 15 meters
13. A tower subtends angles α , 2α and 3α , respectively at points A, B and C which are lying on a horizontal line through the foot of the tower. Then $\frac{AB}{BC}$ is equal to
NIMCET 2025
(a) $1 + 2\cos 2\alpha$ (b) None of these
(c) $2 + \cos 3\alpha$ (d) $2 + 3\cos 3\alpha$
14. A tower subtends an angle of 30° at a point on the same level as the foot of the tower. At a second point h meters above the first, the depression of the foot of the tower is 60° . What is the horizontal distance of the tower from the point?
NIMCET 2025
(a) $2h \cot 60^\circ$ (b) $2h \cot 60^\circ$
(c) $h \tan 60^\circ$ (d) $2h \tan 60^\circ$





15. The angles of depression of the top and bottom of an 8m tall building from the top of a multi-storied building are 30° and 45° , respectively. What is the height of the multistoried building and the distance between the two buildings? **NIMCET 2025**

- (a) $4(3 + \sqrt{3})$ m, $4(3 + \sqrt{3})$ m
 (b) $2(3 + \sqrt{3})$ m, $2(3 + \sqrt{3})$ m
 (c) $4(2 + \sqrt{2})$ m, $4(2 + \sqrt{2})$ m
 (d) $2(3 - \sqrt{3})$ m, $4(3 + \sqrt{3})$ m

16. An airplane, when 4000m high from the ground, passes vertically above another airplane at an instant when the angles of elevation of the two airplanes from the same point on the ground are 60° and 30° , respectively. Find the vertical distance between the two airplanes **NIMCET 2025**

- (a) 16000/7m (b) 8000/3m
 (c) 8000/7m (d) 1200m

ANSWER KEY

1.	b	2.	d	3.	a	4.	d	5.	a
6.	c	7.	b	8.	a	9.	a	10.	a
11.	d	12.	c	13.	a	14.	b	15.	a
16.	b								

Analysis 2007 to 2025

Questions Asked in NIMCET 2007 to 2025

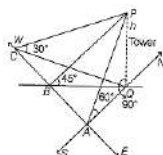
Years	No. of Questions	Years	No. of Questions
NIMCET 2007	00	NIMCET 2016	01
NIMCET 2008	00	NIMCET 2017	00
NIMCET 2009	00	NIMCET 2018	00
NIMCET 2010	01	NIMCET 2019	00
NIMCET 2011	01	NIMCET 2020	01
NIMCET 2012	01	NIMCET 2021	00
NIMCET 2013	01	NIMCET 2022	02
NIMCET 2014	01	NIMCET 2023	01
NIMCET 2015	04	NIMCET 2024	00
		NIMCET 2025	04

SOLUTION

1. (b)

$$AQ = h \cot 60^\circ = \frac{h}{\sqrt{3}}, BQ = h, CQ = \sqrt{3}h$$

From right angled $\triangle ABQ$, $\angle A = 90^\circ$, we have



$$AB^2 = BQ^2 - AQ^2 = \frac{2}{3}h^2 \Rightarrow AC^2 = CQ^2 - AQ^2 = \frac{8}{3}h^2$$

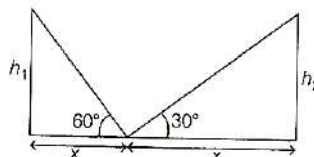
$$\therefore \frac{AC}{AB} = 2$$

$$AC = 2AB \Rightarrow AB + BC = 2AB \Rightarrow AB = BC \Rightarrow \frac{AB}{BC} = 1$$

2 (d) $\tan 60^\circ = \frac{h_1}{x}$

$$\sqrt{3}x = h_1 \quad \dots\dots(i)$$

$$\tan 30^\circ = \frac{h_2}{x} \Rightarrow \frac{1}{\sqrt{3}} = \frac{h_2}{x}$$



$$\Rightarrow h_2 = \frac{x}{\sqrt{3}} \quad \dots\dots(ii)$$

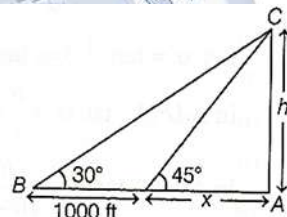
On dividing Eq. (i) by Eq. (ii), we get

$$\Rightarrow \frac{h_1}{h_2} = \frac{\sqrt{3}x}{\frac{x}{\sqrt{3}}} = \frac{\sqrt{3}x(\sqrt{3})}{x} = \frac{3}{1} = 3:1$$

3. (a) $\tan 45^\circ = \frac{h}{x} \Rightarrow h = x$

$$\text{and } \tan 30^\circ = \frac{1}{\sqrt{3}} = \frac{h}{x+1000}$$

$$\Rightarrow x + 1000 = \sqrt{3}h$$



$$\Rightarrow x + 1000 = \sqrt{3}x \quad (h = x) \Rightarrow \frac{1000}{\sqrt{3}-1} = x$$

$$\Rightarrow AB = \frac{1000}{\sqrt{3}-1} + 1000 = 1000 \left(\frac{1+\sqrt{3}-1}{\sqrt{3}-1} \right)$$

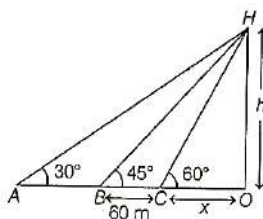
$$= \frac{(\sqrt{3})1000}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = \frac{(\sqrt{3}+1)1000\sqrt{3}}{2}$$

$$= 500\sqrt{3}(\sqrt{3}+1)\text{ft}$$

4. (d) $\tan 60^\circ = \frac{h}{OC}$

$$H = OC \sqrt{3} \Rightarrow h = \sqrt{3}x$$

$$\Rightarrow x + 60 = \sqrt{3}x \Rightarrow x = \frac{60}{\sqrt{3}-1}$$



$$\tan 30^\circ = \frac{h}{AO} = \frac{\sqrt{3}x}{AB+60+x}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{\sqrt{3}(60)}{(\sqrt{3}-1)} \times \frac{1}{AB+60+\frac{60}{\sqrt{3}-1}}$$

$$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}(60)}{AB(\sqrt{3}-1)+\sqrt{3}(60)-60+60}$$

$$1 = \frac{3(60)}{AB(\sqrt{3}-1)+\sqrt{3}(60)}$$

$$\Rightarrow AB(\sqrt{3}-1) + \sqrt{3}(60) = (3)(60)$$

$$\Rightarrow AB = \sqrt{3}(60)$$

$$\Rightarrow AB = 60\sqrt{3} \text{ m}$$



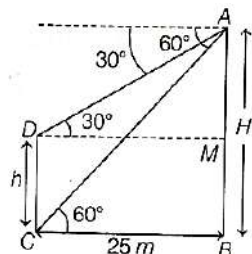
5. (a) $AB = H$ (say)

$$\Rightarrow \tan 30^\circ = \frac{AM}{25} \Rightarrow AM = \frac{25}{\sqrt{3}}$$

$$\Rightarrow \tan 60^\circ = \frac{AB}{25}$$

$$\Rightarrow \sqrt{3} = \frac{AM+MB}{25} = \frac{25/\sqrt{3}+MB}{25}$$

$$\Rightarrow MB = \frac{50}{\sqrt{3}}$$



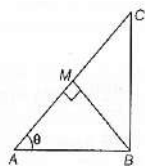
$$\Rightarrow AB = H = AM + MB = \frac{25}{\sqrt{3}} + 25 \left(\frac{2}{\sqrt{3}} \right)$$

$$H = 25\sqrt{3}$$

$$\Rightarrow \text{Height of 2nd tower} = h = CD = MB = \frac{50}{\sqrt{3}}$$

6. (c) $MB = x$ (say)

$$\Rightarrow AC = 4x \Rightarrow \cos \theta = \frac{AB}{4x}$$



In $\triangle AMB$

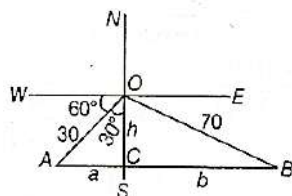
$$\Rightarrow \sin \theta = \frac{MB}{AB} = \frac{x}{AB} \Rightarrow \sin \theta \cdot \cos \theta = \frac{AB}{4x} \cdot \frac{x}{AB} = \frac{1}{4}$$

$$\Rightarrow 2\sin \theta \cos \theta = \frac{2}{4} = \frac{1}{2} \Rightarrow \sin 2\theta = \sin 30^\circ$$

$$\Rightarrow 2\theta = 30^\circ \Rightarrow \theta = 15^\circ$$

7. (b) $\sin 30^\circ = \frac{a}{30} \Rightarrow a = 30 \times \sin 30^\circ = 15 \text{ km}$

$$\cos 30^\circ = \frac{OC}{30}$$



$$\Rightarrow OC = 15\sqrt{3}$$

Now, using Pythagoras theorem in $\triangle OCB$

$$h^2 + b^2 = 4900$$

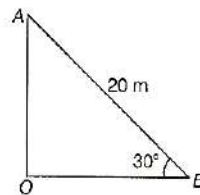
$$\Rightarrow b = \sqrt{4900 - (15\sqrt{3})^2} = \sqrt{4900 - 675} = \sqrt{4225} = 65$$

$$\text{Now, } a + b = 65 + 15 = 80$$

$$\text{So, it will take } t = \frac{80}{10} = 8 \text{ hrs and Noon} = 12:00 \text{ pm}$$

$$\text{So, after 8 hrs, time will be } 8:00 \text{ pm.}$$

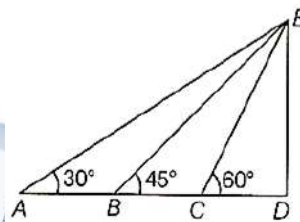
8. (a) $AB = 20 \text{ m}, AO = ?$



$$\sin 30^\circ = \frac{OA}{AB}$$

$$\Rightarrow \frac{1}{2} = \frac{OA}{20} \Rightarrow OA = 10 \text{ m}$$

9.(a)



In $\triangle EAD$

$$\tan 30^\circ = \frac{ED}{AD} = \frac{1}{\sqrt{3}} \Rightarrow AD = \sqrt{3}ED$$

$$\text{In } \triangle EBD, \tan 45^\circ = \frac{ED}{BD} = 1 \Rightarrow ED = BD$$

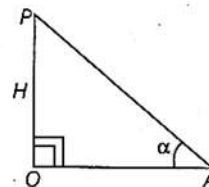
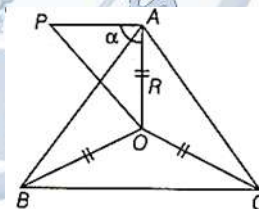
$$\text{In } \triangle ECD, \tan 60^\circ = \frac{ED}{CD} = \sqrt{3} \Rightarrow CD = \frac{ED}{\sqrt{3}}$$

$$\frac{AB}{BC} = \frac{AD-BD}{BD-CD} = \frac{\sqrt{3}-1}{1-\frac{1}{\sqrt{3}}} = \frac{\sqrt{3}(\sqrt{3}-1)}{(\sqrt{3}-1)} = \frac{\sqrt{3}}{1}$$

$$\frac{AB}{BC} = \frac{\sqrt{3}}{1} = \sqrt{3}:1$$

10. (a) Refer to solution 9.

11. (d)

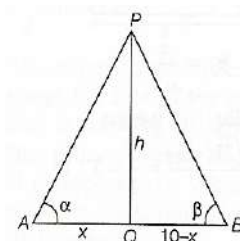


$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R \quad \tan \alpha = \frac{h}{R}$$

$$\tan \alpha = \frac{2H \sin A}{a}$$

$$\Rightarrow H = \frac{1}{2} a \operatorname{cosec} A \tan \alpha$$

12. (c)



$$\text{Let } \alpha = \tan^{-1} 3 \Rightarrow \tan \alpha = 3 \text{ and } \beta = \tan^{-1} 2 \Rightarrow \tan \beta = 2$$

$$\text{In } \triangle APQ, \tan \alpha = \frac{h}{x} = 3 \Rightarrow x = \frac{h}{3}$$

$$\text{In } \triangle BPQ, \tan \beta = \frac{h}{10-x} = 2$$

$$\Rightarrow h = 20 - 2x \Rightarrow h = 20 - \frac{2h}{3}$$

$$\Rightarrow h = 12 \text{ m}$$



13. (a)

A tower subtends angles α , 2α and 3α , respectively at points A, B and C which are lying on a horizontal line through the foot of the tower. Find $\frac{AB}{BC}$.

Let the height of the tower be h

At a point where angle of elevation = θ , distance from tower = $h \cot \theta$.

$$OA = h \cot(\alpha), OB = h \cot(2\alpha), OC = h \cot(3\alpha)$$

Find AB and BC

$$AB = OA - OB = h(\cot \alpha - \cot 2\alpha)$$

$$BC = OB - OC = h(\cot 2\alpha - \cot 3\alpha)$$

Ratio AB/BC

$$\frac{AB}{BC} = \frac{(\cot \alpha - \cot 2\alpha)}{(\cot 2\alpha - \cot 3\alpha)}$$

$$\text{Simplify using } \cot A - \cot B = \frac{\sin(B-A)}{(\sin A \sin B)}$$

$$\cot \alpha - \cot 2\alpha = \frac{\sin(2\alpha - \alpha)}{(\sin \alpha \sin 2\alpha)} = \frac{\sin \alpha}{(\sin \alpha \sin 2\alpha)} = \frac{1}{\sin 2\alpha}$$

$$\cot 2\alpha - \cot 3\alpha = \frac{\sin(3\alpha - 2\alpha)}{(\sin 2\alpha \sin 3\alpha)} = \frac{\sin \alpha}{(\sin 2\alpha \sin 3\alpha)}$$

Final ratio

$$\frac{AB}{BC} = \frac{\frac{1}{\sin 2\alpha}}{\frac{\sin \alpha}{\sin 2\alpha \sin 3\alpha}} = \frac{1}{\sin 2\alpha} \cdot \frac{\sin 2\alpha \sin 3\alpha}{\sin \alpha} = \frac{\sin 3\alpha}{\sin \alpha}$$

Use triple angle identity

$$\sin 3\alpha = 3 \sin \alpha - 4 \sin^3 \alpha$$

$$= \frac{\sin 3\alpha}{\sin \alpha} = 3 - 4 \sin^2 \alpha$$

$$\text{Now use } \sin^2 \alpha = \frac{1 - \cos 2\alpha}{2}$$

$$= 3 - 4 \cdot \frac{(1 - \cos 2\alpha)}{2} = 3 - 2(1 - \cos 2\alpha)$$

$$= 1 + 2 \cos 2\alpha$$

Ans.

$$\frac{AB}{BC} = 1 + 2 \cos 2\alpha$$

14. (b)

$$\text{From A: } \tan 30^\circ = \frac{H}{h} \Rightarrow x = \sqrt{3}H$$

$$\text{From B: } \tan 60^\circ = \frac{h}{x} \Rightarrow x = \frac{h}{\sqrt{3}}$$

$$\text{Equating: } \sqrt{3}H = \frac{h}{\sqrt{3}} \Rightarrow H = \frac{h}{3}$$

$$\tan x = \sqrt{3}H = \frac{h}{\sqrt{3}} = 2h \cot 60^\circ$$

15. (a)

Let distance = x , height of tall building = H .

From 30° (top of small building height 8 m):

$$\tan 30^\circ = \frac{H-8}{x} \Rightarrow x = \sqrt{3}(H-8)$$

From 45° (bottom of small building):

$$\tan 45^\circ = \frac{H}{x} \Rightarrow x = H$$

Equating:

$$H = \sqrt{3}(H-8) \Rightarrow H = 4(3 + \sqrt{3})$$

$$\text{Thus } x = H - 4(3 + \sqrt{3})m.$$

Ans. $4(3 + \sqrt{3})m$ for both height & distance

16. (b)

- Let the height of the upper airplane be $H = 4000m$.
- Let the height of the lower airplane be h .
- Let the horizontal distance from the point on the ground to the point vertically below the airplanes be x . [1]
- The angle of elevation for the upper airplane is 60° , so, $\tan(60^\circ) = \frac{H}{x}$
- The angle of elevation for the lower airplane is 30° , so, $\tan(30^\circ) = \frac{h}{x}$.
- From the equation for the upper airplane, $x = \frac{H}{\tan(60^\circ)} = \frac{4000}{\sqrt{3}}$
- Substitute the value of x into the equation for the lower airplane: $h = x \cdot \tan(30^\circ) = \frac{4000}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}}$
- Calculate the height of the lower airplane: $h = \frac{4000}{3}m$
- The vertical distance between the two airplanes is $D = H - h$.
- Calculate the vertical distance: $d = 4000 - \frac{4000}{3} = \frac{1200-4000}{3} = \frac{8000}{3}m$.

Ans.

$$\frac{8000}{3}m \approx 2666.67m$$